

3.9.1 Telescopes (A-level only)

3.9.1.1 Astronomical telescope consisting of two converging lenses (A-level only)

Content

Ray diagram to show the image formation in normal adjustment.

Angular magnification in normal adjustment.

$$M = \frac{\text{angle subtended by image at eye}}{\text{angle subtended by object at unaided eye}}$$

Focal lengths of the lenses.

$$M = \frac{f_o}{f_e}$$

3.9.1.2 Reflecting telescopes (A-level only)

Content

Cassegrain arrangement using a parabolic concave primary mirror and convex secondary mirror.

Ray diagram to show path of rays through the telescope up to the eyepiece.

Relative merits of reflectors and refractors including a qualitative treatment of spherical and chromatic aberration.

Similarities and differences of radio telescopes compared to optical telescopes. Discussion should include structure, positioning and use, together with comparisons of resolving and collecting powers.

3.9.1.4 Advantages of large diameter telescopes (A-level only)

Content

Minimum angular resolution of telescope.

Rayleigh criterion, $\theta \approx \frac{\lambda}{D}$

Collecting power is proportional to *diameter*².

Students should be familiar with the rad as the unit of angle.

Comparison of the eye and CCD as detectors in terms of quantum efficiency, resolution, and convenience of use.

No knowledge of the structure of the CCD is required.